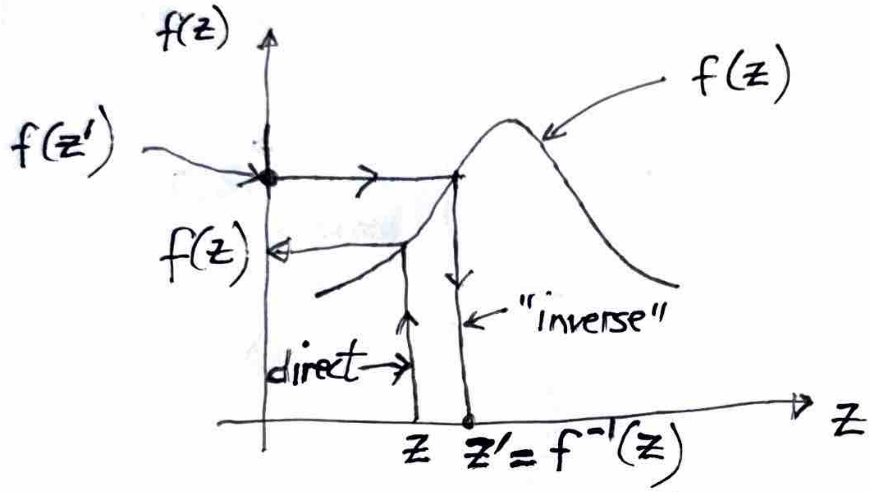
(iii) Inverse problem



$\underbrace{z' = f^{-1}(z)}_{\text{inverse function}}$

Given $F(z=z')$, find $z' = F^{-1}(z)$

example, $F(z') = 0.95$, $z' = F^{-1}(0.95) = 1.65$

6. Back to safety inventory

PAC : If $z \leq z'$, then product is available during the lead-time, L $\longrightarrow$ (7) repeated

$$CSL \triangleq \text{Prob.}(z \leq z') \longrightarrow (8) \text{ repeated}$$

$$\text{Prob.}(z \leq z') = F(z=z') \longrightarrow (9) \text{ repeated}$$

$$\boxed{CSL \underset{(8)}{=} \text{Prob.}(z \leq z') \underset{(9)}{=} F(z=z')} \longrightarrow (10)$$

If we require $CSL = (CSL)_{desired}$, $\longrightarrow (11)$,

then $(CSL)_{desired} \underset{(10,11)}{=} F(z = z')$ $\longrightarrow (12)$

$$\Rightarrow z' = F^{-1}\left[(CSL)_{desired}\right] \longrightarrow (13)$$

$$z' \underset{(6)}{=} \frac{ROP - (D_L)_m}{\sigma_L} \underset{3(c)}{=} \frac{ss}{\sigma_L} \underset{(13)}{=} F^{-1}\left[(CSL)_{desired}\right]$$

$$ss = \sigma_L \, F^{-1}\left[(CSL)_{desired}\right] \longrightarrow (14)$$

$$\boxed{ss \underset{(2)}{=} \sqrt{L}\,\sigma_W \, F^{-1}\left[(CSL)_{desired}\right]} \longrightarrow (15)$$

For our example:

given
$$\begin{cases} (CSL)_{desired} = 0.95 \text{ (or } 95\%) \\ \sigma_W = 200 \text{ units} \\ D_W = 300 \text{ units} \\ L = 2 \text{ weeks} \end{cases}$$

(σ_W and D_W: weekly demand statistics)

$$ss \underset{(15)}{=} \left[\sqrt{2}\,(200)\right] F^{-1}\left[(CSL)_{desired} = 0.95\right]$$

$$= \left[\sqrt{2}\,(200)\right] \underbrace{F^{-1}(0.95)}_{\text{from statistical tables}}$$

$$\Rightarrow \quad SS = \left[\sqrt{2}\,(200)\right](1.65)$$

$$SS = 466 \text{ cell-phones}$$

Re-order point, $ROP \underset{3(b)}{=} L D_W + SS$

$$= (2)(300) + 466$$

$$ROP = 1066 \text{ cell phones}$$

Conclusion (or Recommendation):

For a desired cycle service level, CSL = 95%,
Best Buy needs to set its Re-order point, (ROP),
at 1066 cell phones; and needs to maintain
a safety stock of 466 phones.
(which is Prob. 4 of HW #5, due today!)

TIM 172B, MOT-II, Lecture 14, 2/19/26

Agenda

0. Complete development of safety inventory results; with worked example (HW #5, Prob. 4)

1. Comments on HW # 5

2. Basic results from calculus-based probability theory

3. Application to safety inventory problems

4. Product availability metrics
   (a) CSL, cycle service level
   (b) f_r, fill-rate

5. Calculating CSL & f_r

6. HW #6 has been posted, and is due Thursday, next week

1. Comments on HW # 5

Prob. 6

Case study : 2 "Plantronics" presentation [from TIM170]

- How Plantronics Supply Chain works [K. Hypko]
- IT at Plantronics (Tom Gill, CIO of Plantronics)

Prob. 3 : 2

Aggregation of Cycle Inventory

Case 1 (No aggregation)

Case 2 (Simple Aggregation)

General question : Should Cycle Inventory be aggregation.

Answer : Yes, because lot-sizes are reduced, ("So what"?) → and therefore, cycle inventory costs are reduced.

Prob. 4 : safety inventory : worked out in class today

Prob. 5 : Walmart : "job-interview" question

1. Basic results from calculus-based probability theory.

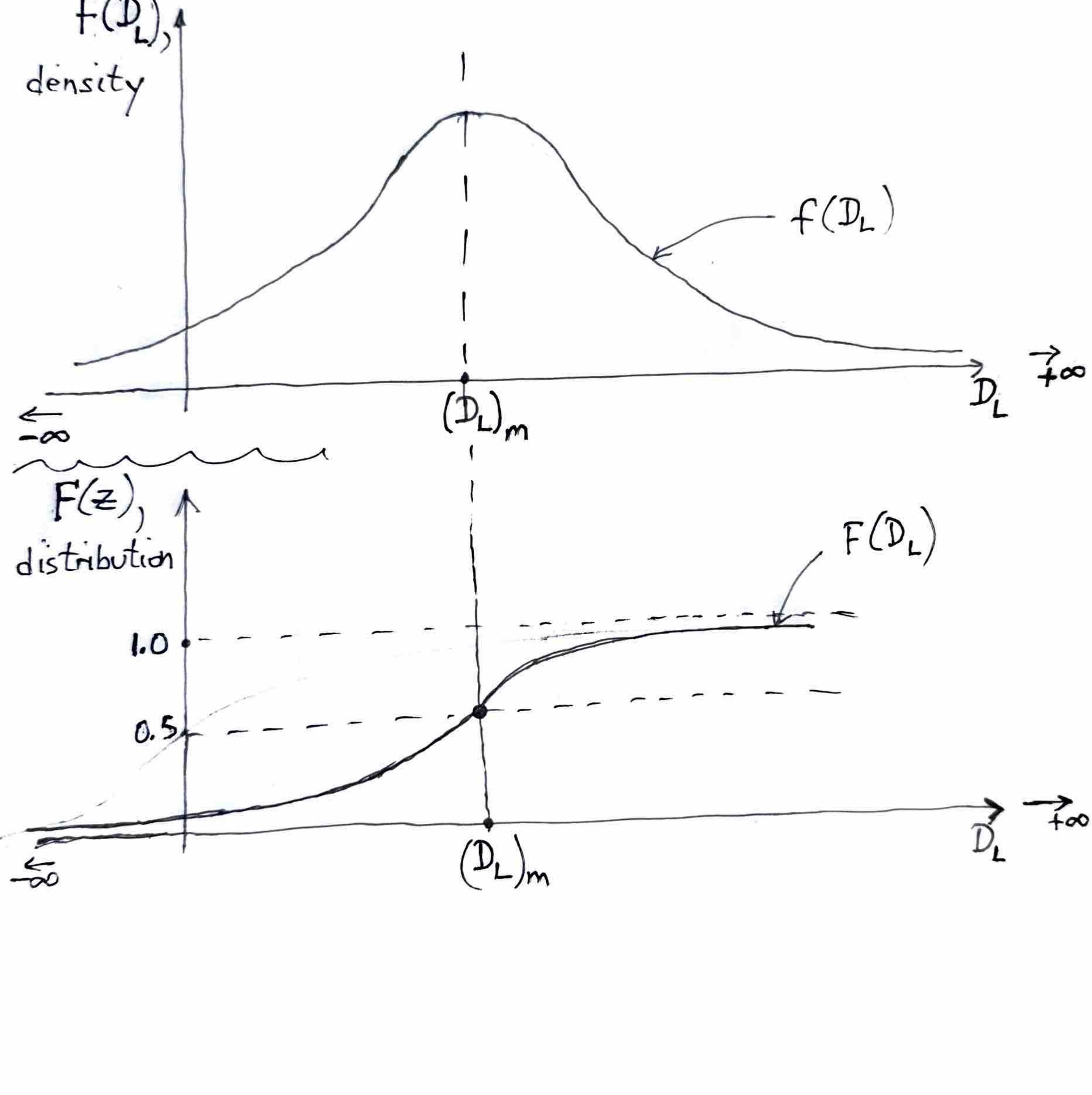
The basic variable of interest for us is D_L, the actual demand during the supplier lead-time, L.

<u>Assumptions</u>: The weekly demand, D_W, and the demand during the lead-time, D_L, are <u>random</u> variables that follow a Gaussian or normal distribution.

There are 2 prob. functions of interest:

$f(D_L)$: prob. density function

$F(D_L)$: prob. distribution or cumulative prob. distribution function
$\Rightarrow$ used to calculate probabilities


f(D_L), density
f(D_L)
(D_L)_m
D_L
→ +∞
← −∞
F(z), distribution
F(D_L)
1.0
0.5
(D_L)_m
D_L
→ +∞
← −∞